

Trevor Nichols

Contact

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01/02/2006
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Career Object

Detail-oriented professional with experience in software development, API endpoint design and implementation, AI training, and frontend design. Deep understanding in backend integration with Microsoft Azure, GitHub actions, and Nix CI/CD. Extensive experience in modeling, animating, and texturing 3D models for commercial purposes. Praised for creative problem-solving, meticulousness, and technical depth. Excels at the integration of computer systems.

Skills

7+ years

- Linux
- Docker
- Podman
- Docker Compose
- Systemd
- Office Suite

6 years

- Node.js
- JS/TS
- Github Actions
- Java
- Nix/Flakes
- C#
- HTML
- CSS
- Python

4 years

- MD Docs
- QEMU

3 years

- .NET
- Azure
- Microsoft Server
- Ubiquiti Unifi
- Docker Swarm
- Quarto

Notable Projects

Bayesian Belief Propagation

<https://github.com/tnichols217/Bayesian-Belief-Propagation>

- **Researched and Implemented** Bayesian Belief Propagation algorithms in TypeScript
- **Applies** relational dynamic graph learning based on any observations

Gerrymandering Simulations

<https://github.com/tnichols217/CSDS133-final>

- **Implemented** a spatial simulation in pure PostgreSQL to simulate counties based on population densities
- **Aggressively Optimized** the code to run in reasonable time in order to perform simulations on multiple states.

Small AI Image Detection

<https://github.com/tnichols217/DSCI453-final-project>

- **Performed a Meta-Analysis** on the training of small neural network models to attempt and learn what models would be able to balance size and accuracy.
- **Optimized and wrote custom parsers** in order to speed up the execution and training speed by multiple factors.

HTMS

<https://github.com/tnichols217/htms>

- **Created and Designed** a custom HTML server side renderer to enable more composable websites built fully without client side JavaScript
- **Integrates with GitHub Actions** to enable easier use for users
- **Built with Nix** for reproducible outputs and compile system

Beta Nu of Theta Chi Bylaws

<https://github.com/beta-nu-theta-chi/bylaws>

- **Designed and Maintain** the CI system for building my fraternity's bylaw website, official documents, and branding
- **Utilized Nix** to ensure reproducible building of the documentation

Personal Experience

Research Contributor - Computer Science

Feb 2025 - Aug 2025

Consultant

Dr. Jolly Bioinformatics Lab | Cleveland, OH, USA

- **Built CI pipelines** to build and use specialized bioinformatics software
 - Utilized both Nix and GitHub Actions
 - Provided support for software that otherwise would be unsupported on modern platforms
 - Provided standardized environments for research and reputability by utilizing Nix
- **Managed System Deployments** for Researcher Workstations
 - Provided Linux support for researchers
 - Assisted in creating specialized tools and scripts to assist with research

Research Contributor - Software Engineer

Sep 2024 - Ongoing

AI4EDU | Cleveland, OH, USA

[github://os-computational-economics/ai4edu-api-service](https://github.com/os-computational-economics/ai4edu-api-service)

- **Built REST backend services** for our AI chatbot using fastAPI
 - Integrating with multiple APIs including OpenAI, Anthropic, and Claude
 - Serves 2 research teams and 5+ classes on campus for over 100 people
 - Provides custom RAG and file access for models for individual teams and groups
- **Led the backend team** of 4 people
 - Enforced good coding practices and styles
 - Review and provide feedback on 100% of code changes
 - Led and implemented over 7 major features

- **Built and maintained databases** using PostgreSQL, DynamoDB, and Pinecone
 - Designed the database schema
 - Maintained and created database migrations
 - Maintain interfaces for various languages across the project
- **Developed workflows and CI/CD** using GitHub workflows to speed up delivery of our product
 - Sped up deployment times by over 75%
 - Created automated PR testing and checks, increasing team efficiency

Facility Assistant

Oct 2023 - Ongoing

Tinkham Veale University Center | Cleveland, OH, USA

- **Worked on small and large teams** to prepare the venue for small and large events

Audio Visual Technician

Sep 2023 - Ongoing

MediaVision | Cleveland, OH, USA

- **Maintained classroom technologies** including both audio and visual systems
- **Facilitated classroom recordings**

Open Source Contributor

Aug 2022 - Ongoing

Contributed to

- NixOS/nixpkgs 19.3k stars
- tnichols217/obsidian-columns 200+ stars, 40k+ downloads (creator)
- Vinzent03/obsidian-git 7.3k stars

Audio Visual Technician

Aug 2020 - Aug 2023

Penang Bible Church | Georgetown, Penang, Malaysia

- **Custom designed an AV system** within the restrictions of a space, needs, and budget
- **Developed preventative measures** to maintain audiovisual systems to reduce breakdowns or malfunctions by over 65%
- **Orchestrated livestreams** and live audio processing for events
- **Designed custom cables** for streamlining connections within an AV system

IT Technician / Programmer Intern

Aug 2022 - Aug 2023

Dalat International School | Tanjung Bungah, Penang, Malaysia

- **Developed backend and frontend software** for identification services including UX design in Aurelia, backend implementation in C#, .NET, Azure functions, and SQL.
- **Developed inventory management systems** for all equipment within several organization's physical and network environments
- In depth knowledge of **Unifi** network management software to manage and maintain a private network

3D Modeler

Nov 2022 - Aug 2023

Sandm. Creative Agency | Georgetown, Penang, Malaysia

- Utilized **GIMP, Inkscape, Illustrator, Photoshop**
 - **Created textures** for 3D models
 - **Generated** bump, metalness, emission maps
- Utilized **Blender**
 - **Modeled** hard surfaces including products, environments, packaging, and buildings
 - **Sculpted** detailed natural shapes whilst maintaining a low poly count
 - Utilized **physics simulations** to animate clothes, smoke, softbodies, and destruction
 - Retopologized models to be **game and animation ready**
 - Highly experienced at **generative geometry**, textures, and lighting
- Led over 50% of 3D design and modelling projects for the company

Education

Case Western Reserve University

Aug 2023 - May 2027

Masters and Bachelors of Science (M.S./B.S.) in Computer Science Candidate

Sec. Bachelors in Data Science, Computer Engineering, Electrical Engineering
Cleveland, OH, US

Senior Status, GPA 3.805

Studied in

- | | |
|--|---|
| <ul style="list-style-type: none"> • Questbridge scholar • University merit-based scholar • Dean's High Honors x4 | <ul style="list-style-type: none"> • Artificial Intelligence • Machine Learning • Data Science Systems and Modelling • Systems Programming • Theoretical Statistics • Hardware Logic Design • Algorithms • Database Systems |
|--|---|

Participated in

- The Fraternity of Theta Chi - Scholarship Chair and Assistant Treasurer since 2024
- CWRUnix (Case Linux Club) - Current President

Dalat International School

Aug 2022 - Aug 2023

High School Diploma with Summa Cum Laude

Penang, Malaysia

GPA: 3.8642

Studied in

- | | |
|--|---|
| <ul style="list-style-type: none"> • 5: AP Computer Science A • 5: AP Computer Science Principles • 5: AP 3D Art and Design • 5: AP Statistics | <ul style="list-style-type: none"> • 5: AP Calculus BC • 5: AP Biology • 5: AP Physics I • 4: AP Music Theory |
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